Lecture 15

So far:

Ramsey — Subgraph

Graph — degree ··· leaf ··· connected — connected component

k-coloring

$\boxed{\chi(G) \leq D+1}$

Path

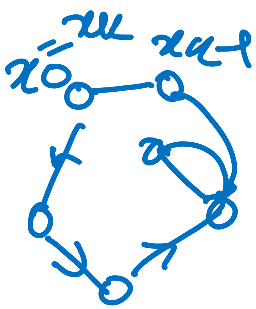
circuit

2-coloring

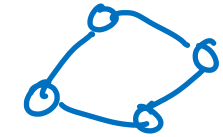
Trees

I Circuits and 2-coloring II Trees III Spanning Trees

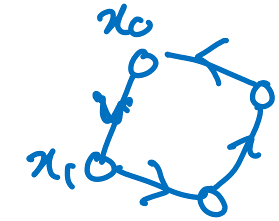
Defn: A path $x_0, e_1, x_1, \ldots, e_k, x_k$ is called a circuit if $x_0 = x_k$. length $= k$.



Remark: A <u>cycle</u> is a graph C_n



<u>circuit</u> is a type of path.



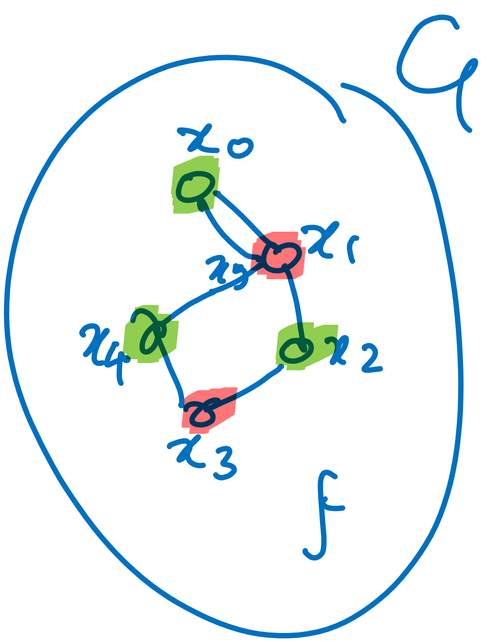
(a circuit has an order, a cycle does not).

Theorem 15.1: A graph is 2-colorable iff it does not contain a circuit of odd length.
($\forall C$ circuit (C has even length))

Proof: $(\longrightarrow)$ Assume $G=(V,E)$ is 2-colorable.

Choose a 2-coloring $f: V \longrightarrow \{1,2\}$.

naming Let $\gamma = x_0, e_1, x_2, e_2, \ldots, e_k, x_{\boxed{k}}$ be a circuit in G.



Consider the sequence of colors
$f(x_0), f(x_1), \ldots, f(x_k)$.

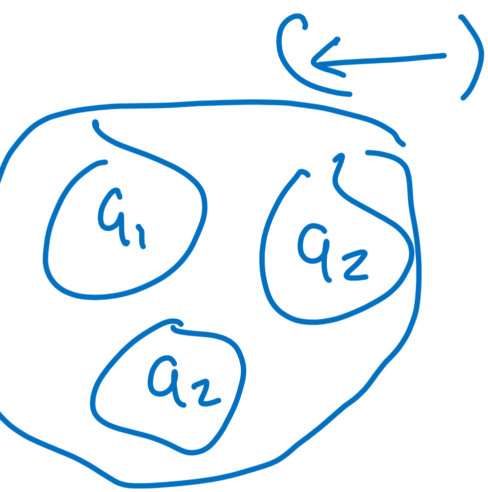
Observe that $f(x_{i+1}) \neq f(x_i)$ for every $i = 0, \ldots, k-1$.

This implies $f(x_i) = f(x_0)$ for every even i

$f(x_i) \neq f(x_0)$ for every odd i.

Since $x_0 = x_k$, we know $f(x_0) = f(x_k)$,

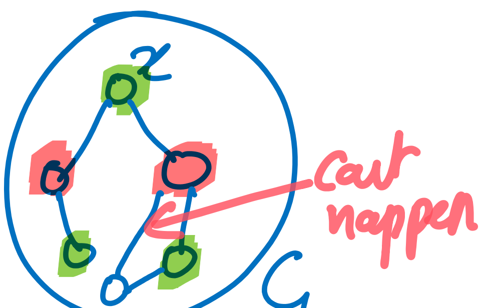
so k must be even, as desired.

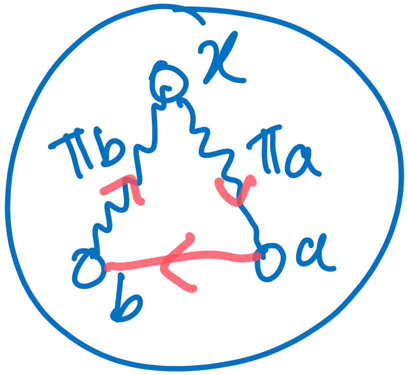


$(\Leftarrow)$ Assume $G = (V, E)$ is connected without loss of generality, by passing to connected components if necessary. Assume G has no circuit of odd length.

Choose an arbitrary vertex $x \in V$.

For every vertex $a \neq x$, choose a path π_a from x to a in G.





Define $f: V \longrightarrow \{1,2\}$ by

$$\begin{cases} f(x) = 1 \\ f(a) = 1 \text{ if } \text{length}(\pi_a) \equiv 0 \pmod 2 \\ f(a) = 2 \text{ if } \text{length}(\pi_a) \equiv 1 \pmod 2 \end{cases}$$

Assume for contradiction that there is some $ab \in E$ such that $f(a) = f(b)$.

By the defn of f,

$$\text{length}(\pi_a) \equiv \text{length}(\pi_b) \pmod 2$$

which implies that

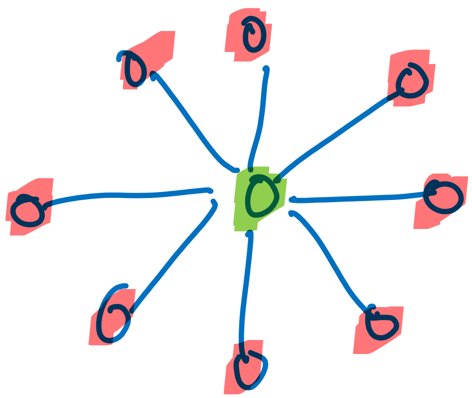
$$\text{length}(\pi_a) + \text{length}(\pi_b) + 1 \equiv 1 \pmod 2.$$

Observe that the circuit

$$C = \pi_a, ab, \text{reversal of } \pi_b$$

is odd, a contradiction. $\square$

<u>ex:</u>

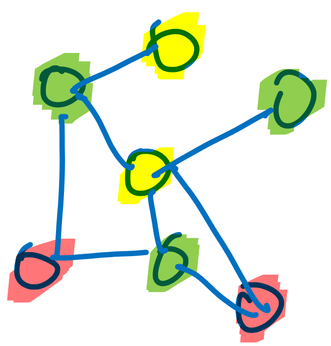


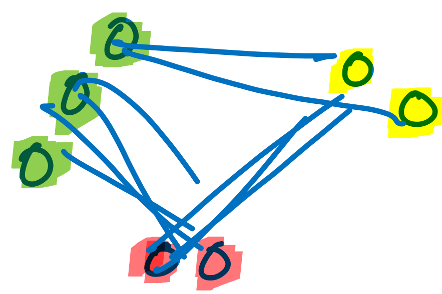
$\Delta = 8$.

Thm 15.1 $\longrightarrow \chi(G) = 2$.

<u>Remark:</u> If $G = (V, E)$ is k-colorable with coloring $f : V \longrightarrow \{1, \dots, k\}$, then for every $i = 1 \dots k$

$f^{-1}(\{i\})$ is an independent set in G.





No edges inside a color class.

II Trees.

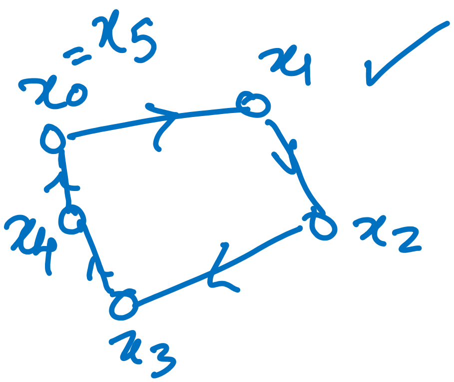
Def: A *cycle* is a circuit with no repeated edges and no repeated vertices except the endpoints, i.e.

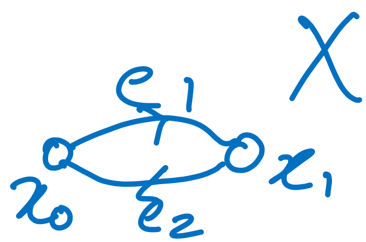
a path $x_0, e_1, x_1, \dots, e_k, x_k$ such that

1. $x_0 = x_k$
2. x_i are distinct for $i \neq 0, k$

$x_i \neq x_0$.

ex:





Not a cycle.

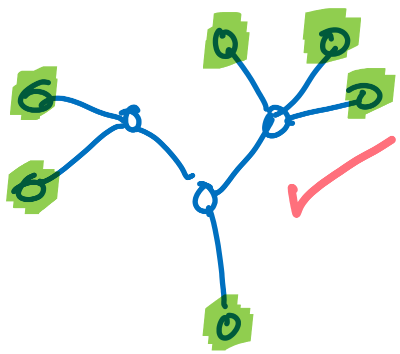
A cycle has ≥ 3 vertices.

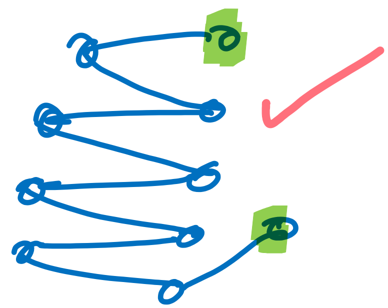
Remark: The set of edges traversed by a cycle forms a cycle graph.

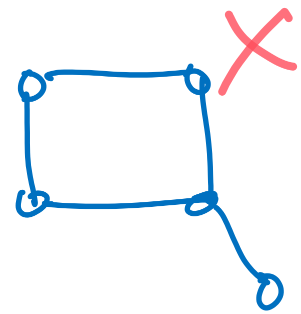
Defn: A graph is called <u>acyclic</u> if it does not contain a cycle. (equivalently, does not contain a cycle graph as a subgraph).

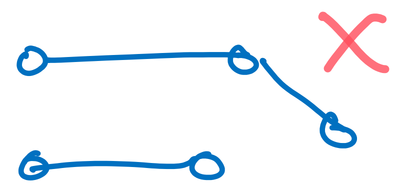
Defn: A <u>tree</u> is a connected acyclic graph.

<u>ex:</u>







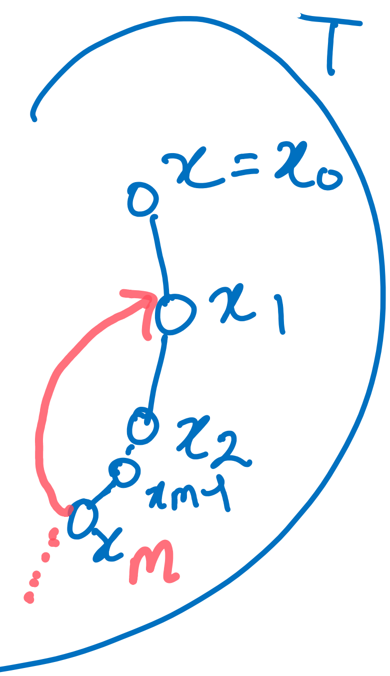


$\emptyset$ ✓

Defn: A <u>leaf</u> in G is a vertex of degree 1.

Theorem 15.2: Every tree with at least 2 vertices has a leaf.

Proof: Let $T=(V,E)$ be an arbitrary tree with at least 2 vertices.



Since T is connected, it has an edge so we can choose a vertex $x \in V$ with degree at least 1.

Recursively define a finite sequence of vertices

$$\begin{cases} x_0 = x \\ x_{k+1} = \text{any vertex adjacent to } x_k \text{ distinct from } x_0, \dots, x_k \text{ if such a vertex exists.} \end{cases}$$

new vertices.

This process terminates in a sequence

$S = x_0, x_1, \dots, x_m$ since G is finite.

Observe that S is a simple path in T.

If x_m is a leaf, we are done.

$\boxed{\deg(x_m) \geq 2}$ If not, x_m must be adjacent to a vertex other than x_{m-1} in S i.e.

name

x_m is adjacent to $\boxed{x_k}$ for $k < m-1$

otherwise we could choose x_{m+1} adjacent to x_m to extend S.

name

Now $x_k, x_{k+1}, \dots, x_m, x_k$ is a cycle in T, a contradiction. $\square$

Corollary: Every tree T with at least 2 vertices has at least 2 leaves.

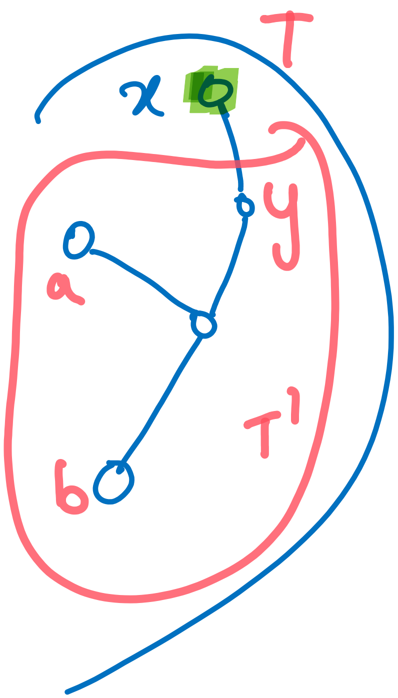
Proof: Proof above shows: For every x of $\deg(x) \geq 1$, T contains a leaf different from x. Applying the same proof to x_M yields a leaf different from x_M. $\square$.

Theorem 15.3: If $T = (V, E)$ tree and $x \in V$ is a leaf, then $T' = (W, F)$ with $[W = V - \{x\},\ F = \{ab \in E : a, b \in W\}]$ is a tree.

$T' = T - x$

Proof: Assume T is a tree and $x \in V$ is a leaf.

Let $T' = T - x$. Since $T' \subseteq T$, T' is acyclic.



Assume a, b are distinct vertices in T'.

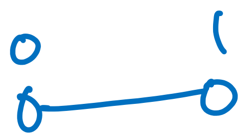
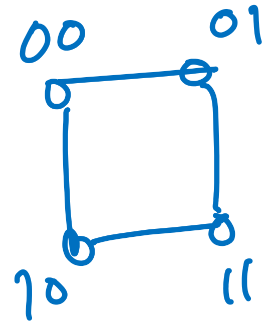
Since T is connected, we can choose a path γ from a to b $\boxed{\text{in } T.}$ not T'!

By HW7.2, we can assume γ is simple.

Notice that γ cannot visit x, since otherwise it would visit $\boxed{\text{the vertex } y}$ name adjacent to x in T twice. Thus, γ is a path in T', as desired. $\square$

HW Tips : $P(n) =$ " . . . holds for all graphs of $\leq n$ vertices."

- H_n

H_1  , H_2  , . . .

Draw Pictures.